Retail Sales Analysis 2025

What drives revenue and profit in a small retail shop, and what should it do next?

This report analyses one year of orders (914 orders, three sales channels, four product categories) with Python, pandas and matplotlib. It answers six business questions and ends with recommendations.

> Portfolio project. **The dataset is synthetic**: it was generated for demonstration and describes no real business. The findings are true for this dataset, and the methods carry over to real data.

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1. Data check

Load the orders and the cost-of-goods (COGS) assumptions, then check for gaps, duplicates and the date range before analysing anything.

	Value
Rows	914
Missing values	0
Duplicate order IDs	0
First date	2025-01-01
Last date	2025-12-31
Non-positive units or prices	0
Categories without COGS %	0

	OrderID	Date	Channel	Category	Customer	Units	UnitPrice	Revenue	COGS	GrossProfit	Mor
0	ORD-0001	2025-01-01	Online	Events & weddings	Returning	1	466.26	466.26	186.50	279.76	
1	ORD-0002	2025-01-01	Phone	Bouquets	New	1	74.28	74.28	33.43	40.85	
2	ORD-0003	2025-01-01	Online	Gifts & add-ons	Returning	1	28.13	28.13	15.47	12.66	
3	ORD-0004	2025-01-01	In-store	Gifts & add-ons	New	2	22.87	45.74	25.16	20.58	
4	ORD-0005	2025-01-02	In-store	Bouquets	New	1	62.05	62.05	27.92	34.13	

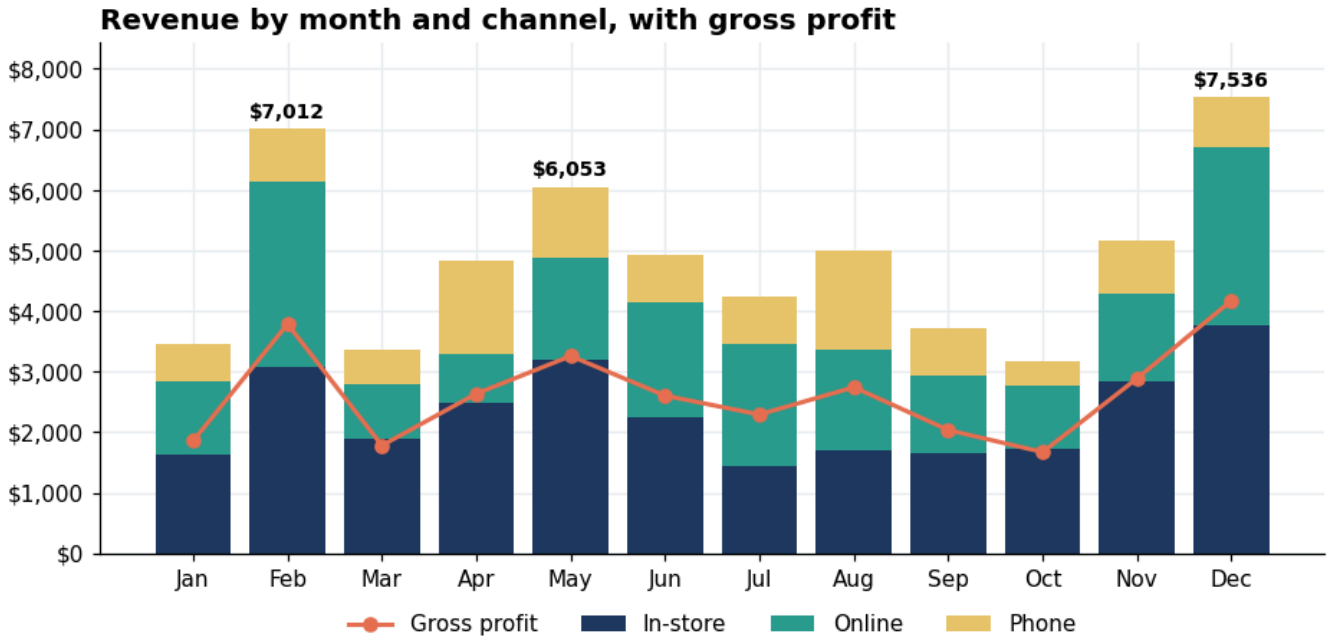
The data is clean: one row per order, no gaps or duplicates, every date inside 2025, every category has a cost-of-goods %. Revenue = units × unit price; gross profit = revenue − COGS.

2. Headline numbers

	Value
Revenue	$58,482.14
Gross profit	$31,712.54
Gross margin	54.2%
Orders	914
Avg order value	$63.98
Median order value	$48.37
Returning customers	45.6%

The **average** order is $63.98 but the **median** is only $48.37: a small number of large orders pulls the average up. Section 4 shows where they come from.

3. Seasonality: when does the shop earn?



Month	Revenue	GrossProfit	Orders	Avg order	vs previous month
Jan	$3,452	$1,856	63	$54.79	–
Feb	$7,012	$3,785	110	$63.74	+103%
Mar	$3,354	$1,780	59	$56.85	-52%
Apr	$4,824	$2,635	73	$66.08	+44%
May	$6,053	$3,258	108	$56.05	+25%
Jun	$4,931	$2,603	86	$57.34	-19%
Jul	$4,240	$2,291	56	$75.71	-14%
Aug	$5,008	$2,742	80	$62.60	+18%
Sep	$3,723	$2,034	56	$66.47	-26%
Oct	$3,182	$1,670	60	$53.03	-15%
Nov	$5,168	$2,887	63	$82.03	+62%
Dec	$7,536	$4,172	100	$75.36	+46%

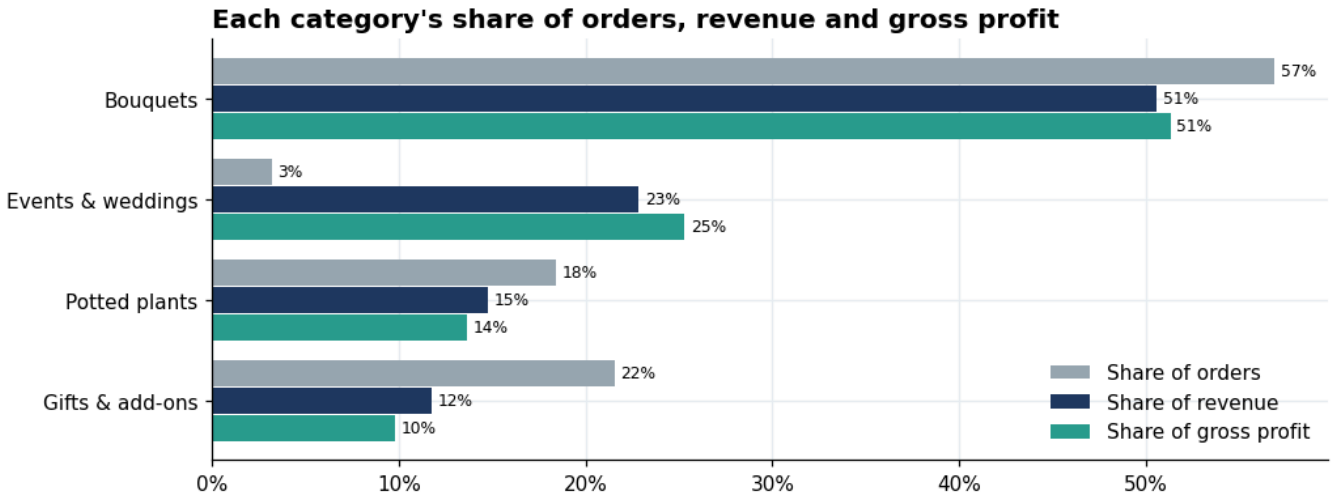
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Feb + May + Dec = 35.2% of annual revenue
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	Share of the month's revenue
Valentine's week (Feb 7–14) in February	47%
Mother's Day week (May 5–11) in May	43%
Pre-Christmas (Dec 15–24) in December	49%

Findings

- Three months, **February, May and December, bring 35% of the year's revenue.** They line up with Valentine's Day, Mother's Day and the holidays.
- The peaks are short: **Valentine's week alone is 47% of February**, Mother's Day week is 43% of May, and the ten days before Christmas are 49% of December.
- February revenue doubles January (+103%) and then halves in March (−52%). October is the weakest month ($3,182).
- Gross profit follows revenue closely: the margin stays in a narrow band all year.

4. Categories: what earns the profit?



Category	Revenue	GrossProfit	Orders	Avg units	Margin	Avg order
Bouquets	29592.290000	16275.759500	520	1.00	0.550000	56.908250
Events & weddings	13363.610000	8018.166000	29	1.00	0.600000	460.814138
Potted plants	8636.110000	4318.055000	168	1.40	0.500000	51.405417
Gifts & add-ons	6890.130000	3100.558500	197	1.42	0.450000	34.975279

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Orders of $300+: 27 (3.0% of orders), 21.9% of revenue; categories: ['Events & weddings']
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Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Event orders	1	3	1	3	2	1	2	3	3	0	4	6

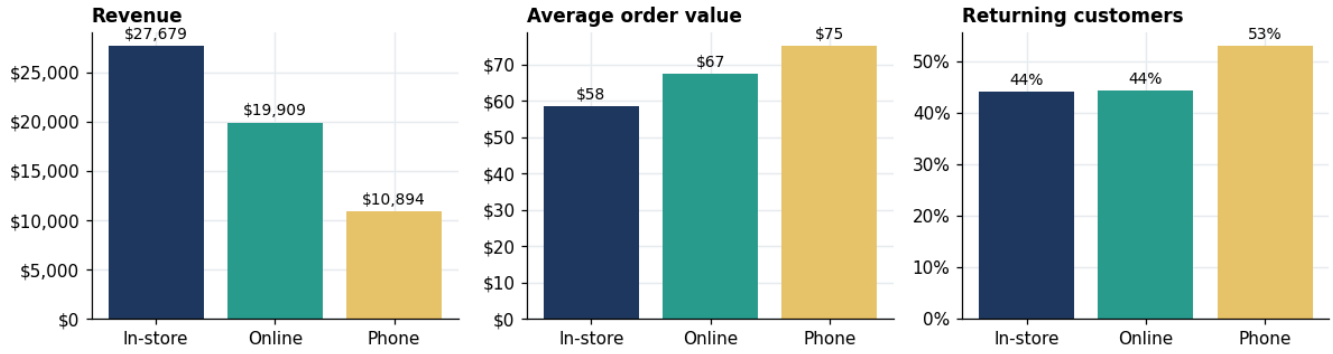
Findings

- **Events & weddings** are only **3% of orders but 23% of revenue and 25% of gross profit**, with the best margin (60%) and an average order of $461. All 27 orders above $300 are events, and they alone bring 22% of revenue.

- That income is lumpy: there were **no event orders at all in October**, which is the main reason it is the weakest month.
- **Bouquets** are the core business: 57% of orders and about half of revenue and profit.
- **Gifts & add-ons** are 22% of orders but only 12% of revenue and 10% of profit, with the lowest margin (45%) and a $35 average order. They work as small add-ons, not as a product line of their own.

5. Channels: how do in-store, online and phone differ?

Sales channels compared



Channel	Revenue	GrossProfit	Orders	Margin	Avg order	Returning share	Events share of revenue
In-store	$27,679	$14,890	474	53.8%	$58.39	44.1%	14.8%
Online	$19,909	$10,851	295	54.5%	$67.49	44.4%	26.2%
Phone	$10,894	$5,971	145	54.8%	$75.13	53.1%	37.1%

Findings

- **In-store** brings the most revenue (47%) with the smallest orders ($58 on average): mostly everyday bouquets.
- **Phone** is the smallest channel (19% of revenue) but has the **highest average order ($75)** and the **most returning customers (53%)**. 37% of its revenue is events: people call to order big arrangements.
- **Online** sits in between, with an average order of $67 and a quarter of its revenue from events.
- Margins are almost the same in every channel (54–55%), so the channels differ in *what* people buy, not in pricing.

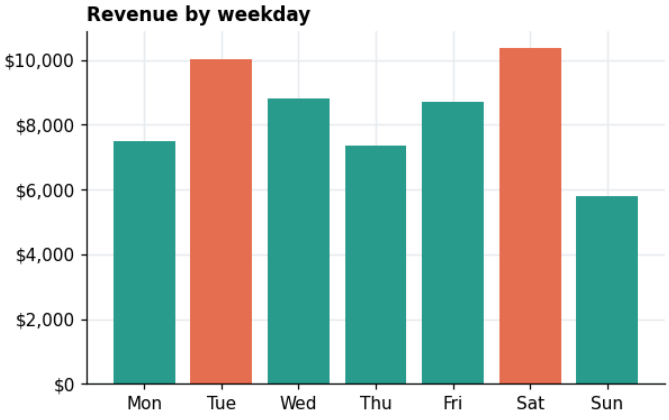
6. Customers: new vs returning

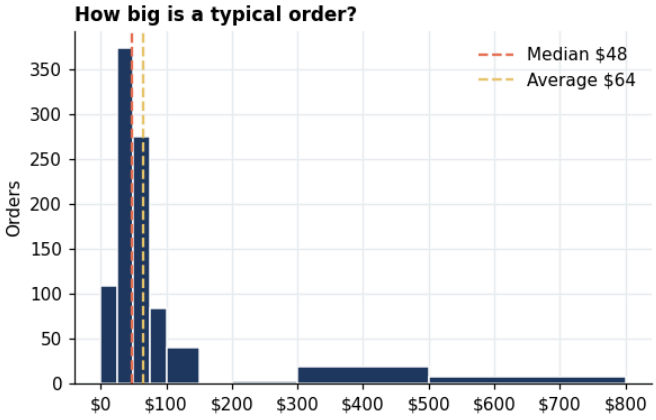
Customer	Revenue	Orders	Avg order	Avg units	Share of orders	Share of revenue
New	33052.580000	497	66.504185	1.21	0.543764	0.565174
Returning	25429.560000	417	60.982158	1.11	0.456236	0.434826

Findings

- **Returning customers place 46% of orders and bring 43% of revenue.** Their average order ($61) is a little below new customers' ($67).
- So loyalty is already strong: almost every second order is a repeat. The opportunity is to make repeat orders bigger, for example with an add-on offer.

7. Weekdays and order size





Weekday	Revenue	Orders	Avg order
Mon	$7,493	135	$55.50
Tue	$10,026	145	$69.14
Wed	$8,798	146	$60.26
Thu	$7,334	111	$66.08
Fri	$8,693	144	$60.37
Sat	$10,360	155	$66.84
Sun	$5,779	78	$74.09

Findings

- **Saturday and Tuesday are the strongest days** (about $10,000 each over the year). **Sunday is the quietest** (78 orders, half of Saturday) but has the **largest average order ($74)**.

- Most orders are small: three quarters are under $66. The long tail on the right is the event orders from section 4.

8. Recommendations

Based on this dataset:

1. **Plan for three short peaks.** Valentine's week, Mother's Day week and the ten days before Christmas bring a large part of the year's revenue. Stock, staff and marketing should be booked for those weeks, with pre-orders opened two to three weeks before.
2. **Grow the events business on purpose.** Events are 3% of orders and a quarter of gross profit, and they fill the quiet months. A clear event package page, a quote form and follow-ups for weddings in the autumn could reduce gaps like October's.
3. **Sell gifts as add-ons to bouquets.** On their own they are small and low-margin. Offered at checkout (card + chocolates, vase), they raise the average bouquet order.
4. **Look after phone customers.** They spend the most per order and come back most often. A callback list and early access to holiday pre-orders suit this group.
5. **Make repeat orders bigger.** Returning customers already make 46% of orders. A small loyalty perk tied to a minimum order value targets their slightly smaller basket.
6. **Check Sunday opening hours.** Sunday has the fewest orders but the largest baskets. Comparing staffing cost with Sunday gross profit would show whether shorter hours or a Sunday-only offer pays off.

Method note: gross profit uses a fixed cost-of-goods % per category. With real data, the next steps would be comparing with the previous year, checking stock-outs and testing the add-on offer before rolling it out.